

Ribosome-Mediated Specificity in Hox mRNA Translation and Vertebrate Tissue Patterning

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Evidence abstract

Historically, the ribosome has been viewed as a complex ribozyme with constitutive rather than regulatory capacity in mRNA translation.

Here we identify mutations of the Ribosomal Protein L38 (Rpl38) gene in mice exhibiting surprising tissue-specific patterning defects, including pronounced homeotic transformations of the axial skeleton.

In Rpl38 mutant embryos, global protein synthesis is unchanged; however the translation of a select subset of Homeobox mRNAs is perturbed.

Our data reveal that RPL38 facilitates 80S complex formation on these mRNAs as a regulatory component of the ribosome to confer transcript-specific translational control.

We further show that Rpl38 expression is markedly enriched in regions of the embryo where loss-of-function phenotypes occur.

Unexpectedly, a ribosomal protein (RP) expression screen reveals dynamic regulation of individual RPs within the vertebrate embryo.

Collectively, these findings suggest that RP activity may be highly regulated to impart a new layer of specificity in the control of gene expression and mammalian development.

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Evidence checklist

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Source continuity

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